

# JAKUB BIAŁAS

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## PROFILE

Graduate in Data Science and Bioinformatics with experience in medical data analysis, statistical modeling, survival analysis, and machine learning. Project work includes oncology, data augmentation, and predictive model evaluation. Conducts experiments and data analyses in R and Python (pandas, numpy, scikit-learn).

Interested in roles such as Data Analyst, Junior Data Scientist, or Bioinformatics Analyst.

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## SKILLS

- **Programming languages:** Python (pandas, numpy, scikit-learn, data analysis, machine learning), R (statistical analysis, survival analysis)
- **Databases:** PostgreSQL, MySQL
- **Data analysis and visualization:** basic knowledge of Power BI and Tableau
- **Tools and environment:** basic knowledge of Linux, Jupyter Notebook, and Git
- **Languages:** Polish (native), English (B2)

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## EDUCATION

- **02/2024 – 09/2025**

**Silesian University of Technology, Faculty of Automatic Control, Electronics and Computer Science**  
**Master's degree, Automation, Electronics and Computer Science, specialization: Data Science -**

Scope: optimization methods, statistics, artificial intelligence, machine learning, data analysis, data visualization.

- **10/2020 – 02/2024**

**Silesian University of Technology, Faculty of Automatic Control, Electronics and Computer Science**  
**Bachelor's degree in Engineering, Biotechnology, specialization: Bioinformatics -**

Scope: bioinformatics, biochemistry, cell biology, genetics, biochemical processes, laboratory and analytical methods.

- **09/2017 – 06/2020**

**IV General Secondary School named after Maria Skłodowska-Curie in Chorzów** – Graduated with distinction. Advanced subjects: mathematics, chemistry, biology.

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## ACHIEVEMENTS, PROJECTS

- **09/2025 – Master's thesis entitled "Augmentation of medical data on response to cancer treatment in small clinical trials"**

- Compared 8 survival data augmentation methods, including bootstrap, SMOTE, Cox model-based approaches, and an original Cox-SMOTE method.

- Developed an original Cox-SMOTE method for generating synthetic survival data for small clinical trials.

- Built an experimental framework to evaluate methods using TES,  $\Delta$ RMST, Spearman and Pearson correlations, variance analysis, and regression parameters.

- Conducted 10 repetitions for each augmentation scenario to assess the stability and reproducibility of results.

- Demonstrated that methods accounting for the structure of survival data, particularly Cox-SMOTE, improve the relationship between TES and  $\Delta$ RMST.

- **01/2024 – Engineering project entitled "Modification and extension of algorithms for estimating spatial heterogeneity of lymphocyte maps in cancer prognosis"**

- Developed a GLCM-based histopathological image analysis algorithm for cancer prognosis.

- Designed an image preprocessing pipeline including filtering, morphological operations, background removal, image cropping, and grayscale level adjustment.

- Evaluated the impact of parameter modifications on prediction quality using the Cox model, Harrell's C-index, and p-value.

- Identified configurations that improved model effectiveness; in the best case, increased the C-index from 0.72 to 0.91.

- **07/2023 – 08/2023 – Student internship at the Klimzowiec wastewater treatment plant**

- Monitored wastewater treatment plant operating parameters in the SCADA system.

- Supported laboratory processes related to sample handling.

- Helped prepare assumptions for the renovation of a technological tank.

- **06/2022 – Cambridge First Certificate (FCE) certificate**

*Jakub Białas*